REACTION WHEEL ATTITUDE CONTROL SYSTEM FOR NANO SATELLITES

A Project Report Submitted to

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal towards the partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering

[SESSION 2025-26]

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A Govt. Aided Autonomous Institute, Affiliated to RGPV, Bhopal

DEPARTMENT OF MECHANICAL ENGINEERING

[Session 2025-26]

RECOMMENDATION

We are pleased to recommend that the project work entitled **Reaction Wheel Attitude Control System for Nano Satellites** submitted by **Arnav Shinde, Ishan Sabnis, Prakhar Shrivastava, Pranav Dhoke and Shubham Pareek** may be accepted in partial fulfillment of the degree of **Bachelor of Technology in Mechanical Engineering** of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the [**session**].

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CERTIFICATE

This is to certify that the Project entitled **Reaction Wheel Attitude Control System for Nano Satellites** submitted by **Arnav Shinde, Ishan Sabnis, Prakhar Shrivastava, Pranav Dhoke and Shubham Pareek** is accepted in partial fulfillment of the degree of **Bachelor of Technology in Mechanical Engineering** of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal during the **Session 2025-26.**

DECLARATION

We **(Arnav Shinde, Ishan Sabnis, Prakhar Shrivastava, Pranav Dhoke and Shubham Pareek from Department of Mechanical Engineering)** declare that the Project **Reaction Wheel Attitude Control System for Nano Satellites** is our own work conducted under the supervision of **Prof. Vinod Parashar from Department of Mechanical Engineering (SGSITS, Indore).**

We affirm that this work is entirely original and has not been copied from any source. All sources consulted during the research process have been duly acknowledged and cited in the bibliography.

We further declare that to the best of our knowledge, this project work does not contain any part of any work which has been submitted for the award of any degree or any other work either in this University or in any other University/ websites without proper citation.

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ABSTRACT

Nano satellites, particularly CubeSats, have gained significant importance in modern space missions due to their compact size, low cost, and rapid development capabilities. However, one of the major challenges associated with nano satellites is achieving precise attitude control under strict constraints of size, mass, and limited power availability. Efficient orientation control is essential for mission-critical operations such as communication, Earth observation, and power generation.

To address this challenge, this project proposes a Reaction Wheel-based Attitude Control System (ACS), which provides a compact and efficient solution for controlling satellite orientation. The system operates on the principle of conservation of angular momentum, where the rotation of a flywheel generates an equal and opposite reaction torque on the satellite, enabling precise attitude adjustments.

In this project, a reaction wheel system consisting of a flywheel, electric motor, and supporting structure has been designed. The flywheel is dimensioned to achieve sufficient moment of inertia while maintaining low mass and compact size. A 3D CAD model of the complete assembly has been developed using Creo Parametric 11.0 to ensure proper alignment, balance, and structural feasibility. Additionally, the system design considers integration with control components such as microcontrollers, sensors, and motor drivers for future implementation.

The proposed system aims to provide smooth, continuous, and accurate attitude control while satisfying the constraints of nano satellites. The design demonstrates feasibility and scalability, making it suitable for further development into a functional prototype or simulation-based validation.

LIST OF FIGURES

LIST OF SYMBOLS

I_s Moment of inertia of satellite ($kg \cdot m^2$)
I_{rw} Moment of inertia of reaction wheel ($kg \cdot m^2$)
ω_s Angular velocity of satellite (rad/s)
ω_{rw} Angular velocity of reaction wheel (rad/s)
α_w Angular acceleration of reaction wheel (rad/s^2)
T Torque ($N \cdot m$)
ρ Density of material (kg/m^3)
R Radius of reaction wheel (m)
M Mass of reaction wheel (kg)
t Thickness of reaction wheel (m)
I_{xx}, I_{yy}, I_{zz} Principal moments of inertia ($kg \cdot m^2$)

LIST OF ABBREVIATIONS

RWACS Reaction Wheel Attitude Control System
ACS Attitude Control System
ADCS Attitude Determination and Control System
MBD Multi-Body Dynamics
FEA Finite Element Analysis
CAE Computer-Aided Engineering
CAD Computer-Aided Design
IMU Inertial Measurement Unit
BLDC Brushless DC Motor
ESC Electronic Speed Controller
PSD Power Spectral Density
GEVS General Environmental Verification Specification
COTS Commercial Off-The-Shelf
RPM Revolutions Per Minute

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1. INTRODUCTION

Nano satellites are a class of small satellites with a mass typically ranging from 1 kg to 10 kg. Among them, CubeSats have emerged as one of the most widely adopted platforms due to their standardized structure (typically 10 × 10 × 10 cm units), low development cost, and shorter development cycles. These satellites are extensively used in applications such as Earth observation, communication systems, scientific experiments, and technology demonstration missions.

The rapid growth of nano satellite missions is driven by advancements in miniaturized electronics and the availability of commercial off-the-shelf (COTS) components. However, despite their advantages, nano satellites operate under strict constraints in terms of size, mass, and power. Typically, a CubeSat generates less than 3 W of power, which significantly limits the design and implementation of onboard subsystems.

One of the most critical subsystems of any satellite is the Attitude Determination and Control System (ADCS). The primary function of ADCS is to control and maintain the orientation (attitude) of the satellite in space. Proper orientation is essential to ensure that antennas are directed toward ground stations, sensors are accurately aligned with targets, and solar panels receive maximum sunlight for power generation.

In the space environment, satellites are subjected to various external disturbances such as gravity gradient forces, atmospheric drag (in low Earth orbit), and solar radiation pressure. These disturbances can cause unwanted rotation and instability. Without an effective attitude control system, the satellite may lose its orientation, leading to mission failure.

Traditionally, large satellites use complex and expensive attitude control systems. However, such systems are not suitable for nano satellites due to their size, cost, and power limitations. Therefore, compact and efficient control mechanisms are required.

Among the various attitude control methods, reaction wheels are widely used in nano satellites due to their ability to provide precise, smooth, and continuous control. A reaction wheel is a rotating mass driven by an electric motor. By accelerating or decelerating the wheel, torque is generated, which changes the orientation of the satellite based on the principle of conservation of angular momentum.

With the increasing use of nano satellites in modern space missions, there is a growing demand for efficient and reliable attitude control systems. However, designing such systems for nano satellites presents several challenges.

Conventional attitude control systems used in larger satellites are often bulky, expensive, and consume significant power. These characteristics make them unsuitable for nano satellite applications, where strict constraints on size, mass, and power must be satisfied.

Additionally, earlier nano satellites relied on passive control methods, which offer limited accuracy and cannot meet the requirements of advanced missions such as high-resolution imaging and precise communication.

Therefore, there is a need to develop a compact, lightweight, and energy-efficient attitude control system that can provide accurate and reliable orientation control within the constraints of nano satellites.

The primary objective of this project is to design and analyze a reaction wheel-based attitude control system suitable for nano satellite applications.
The specific objectives of the project are as follows:

- To study the fundamentals of attitude control systems used in nano satellites
- To understand the working principle of reaction wheels based on conservation of angular momentum
- To design a compact reaction wheel mechanism (flywheel) capable of generating sufficient control torque
- To select appropriate motor and structural components for efficient operation
- To develop a CAD model of the reaction wheel assembly using suitable software tools
- To analyze design parameters such as size, weight, and feasibility within nano satellite constraints

❖ Basics of Attitude Control

Attitude Determination and Control Systems (ADCS) for nanosatellites frequently utilize reaction wheels as primary actuators to achieve high-precision pointing and stabilization. These devices function as momentum exchange actuators, leveraging the principle of conservation of angular momentum: by accelerating an internal motorized flywheel, an equal and opposite torque is exerted on the satellite body, thereby altering its orientation. For comprehensive 3-axis control, reaction wheels are typically arranged in an orthogonal or tetrahedral configuration to manage the satellite's primary degrees of freedom: roll, pitch, and yaw.

In a standard 3-axis stabilized mission, the reaction wheels are aligned with the satellite's principal axes of inertia. Adjustments to the flywheel's angular velocity generate the necessary control torques to correct errors in roll (rotation about the longitudinal axis), pitch (rotation about the lateral axis), and yaw (rotation about the vertical axis). Because external environmental disturbances, such as atmospheric drag and solar radiation pressure, can cause a persistent build-up of angular momentum, reaction wheels may eventually reach their maximum rotational speed, a state known as saturation. To recover control capability, nanosatellites often employ secondary actuators, such as magnetorquers, to perform momentum dumping—desaturating the wheels by transferring excess momentum to the Earth's magnetic field.

- ❖ Reaction Wheel Concept

The operational foundation of a reaction wheel is based on the Law of Conservation of Angular Momentum. In the isolated environment of space, the total angular momentum of a nanosatellite remains constant unless acted upon by an external torque. By integrating a motorized flywheel—the reaction wheel—into the satellite's internal structure, the system can manipulate its orientation by redistributing momentum between the wheel and the satellite body. When the internal motor applies a torque to increase the flywheel's spin velocity, the satellite body experiences an equal and opposite reactive torque. This fundamental interaction allows for precise control over the spacecraft's attitude without the need for expendable propellants.

Basically, total angular momentum of the whole satellite + reaction wheels system is conserved.

$$I_S \text{ x } \omega_S + I_{rw} \text{ x } \omega_{rw} = 0$$

$$I_{rw} = - (I_S \text{ x } \omega_S)/ \omega_{rw}$$

This negative means that the reaction wheels are to be rotated in negative sense as that of the satellite. To maintain stability across all three spatial dimensions, a nanosatellite typically employs a minimum of three wheels aligned with the roll, pitch, and yaw axes. By precisely modulating the speed of each wheel, the control system can counteract internal vibrations or rotate the vehicle to a specific target orientation.

2. LITERATURE REVIEW

The growing proliferation of nano-satellites — small spacecraft typically weighing between 1 kg and 10 kg — has transformed the landscape of modern space exploration and Earth observation. Despite their compact form, these platforms must perform highly precise orientation tasks such as Earth imaging, inter-satellite communication, and scientific data acquisition — all of which depend critically on accurate Attitude Determination and Control Systems (ADCS). Reaction wheels are the most widely employed actuators for three-axis attitude control in nano-satellites. The theoretical backbone of any reaction wheel system is the conservation of angular momentum. Markley and Crassidis (2014), in their authoritative text Fundamentals of Spacecraft Attitude Determination and Control, provide a comprehensive mathematical framework describing how reaction wheels exchange angular momentum with a spacecraft body. As the wheel accelerates or decelerates, Newton's Third Law dictates a counter-rotation of the satellite proportional to the inertia ratio between the wheel and the spacecraft.

Wertz (1978) in Spacecraft Attitude Determination and Control expanded this foundation to multi-axis systems, demonstrating that a minimum of three reaction wheels — mounted along orthogonal axes — are required for full three-axis control. He formalised how external torques alter the total system angular momentum while internal torques (such as wheel spin-up) redistribute momentum between the wheel and spacecraft without net external effect.

The mechanical design of the flywheel is the most direct determinant of a reaction wheel's angular momentum storage capability and torque output. The primary design problem is the optimisation of the flywheel geometry to maximise the inertia-to-mass ratio within the geometric envelope imposed by the satellite form factor. Oland and Schlanbusch (2009), in their paper Reaction Wheel Design for CubeSats, present a systematic sizing methodology applicable to CubeSat missions. They derive design equations for customising reaction wheels to different CubeSat sizes and show the relationship between flywheel dimensions and the achievable angular momentum. Their work identifies the operational altitudes at which magnetic torquers are suitable for momentum dumping — a consideration directly linked to flywheel saturation design.

From a commercial design perspective, research from Khalifa University (2023) presents a miniaturised flight-model reaction wheel only 33 mm in diameter and weighing under 38 g, demonstrating that COTS brushless DC motors paired with appropriately sized flywheels can meet CubeSat ACS requirements. Their work includes a torque-versus-diameter plot mapping state-of-the-art reaction wheel models, indicating that wheels below 6 cm diameter are primarily suited to CubeSat applications, while larger satellites require larger flywheels and more powerful motors.

PID-Based Control as Standard Approach
Most nanosatellite RWACS implementations use PID controllers due to simplicity and

reliability. Studies like Narkiewicz et al. (2020) and Long (2014) demonstrate successful implementation across detumbling, nominal control, and momentum unloading, including real hardware validation with IMU and sensor fusion.

Multi-Loop and Structured PID Architectures
Advanced PID configurations such as dual-loop control (inner loop for wheel speed, outer loop for attitude) improve system stability. Mrzljak et al. (2021) validated this using transfer function modelling, Bode plots, and step response analysis for all three axes.

Sensor-Based Feedback and Actuation Loop
Gyroscopic sensors (IMUs) are used to measure angular velocity and detect disturbance-induced attitude errors. This data is processed by an onboard microcontroller, which computes the control action (e.g., via PID) and sends torque commands to the reaction wheel motors to generate stabilizing motion. Long (2014) experimentally validates this closed-loop architecture using IMU feedback, while standard ADCS design frameworks (e.g., Wertz, *Spacecraft Attitude Determination and Control*) describe this sensor–controller–actuator loop as the core of satellite attitude stabilization.

Optimized and Intelligent Control Techniques
Researchers have enhanced PID performance using optimization methods like Genetic Algorithms (Mohamed et al., 2022), achieving better rise time and reduced steady-state error. Additionally, fuzzy logic controllers are used for robustness against uncertainties and efficient momentum management.

3. PROBLEM FORMULATION

❖ Context: The Role of Precision Attitude Control

Reaction wheels (RW) serve as the primary actuators for high-precision, propellant-free attitude control in modern satellite missions. By modulating the angular momentum of an internal flywheel, these systems generate predictable reaction torques that enable three-axis stabilization. This capability is critical for missions requiring high "pointing stability," such as Earth observation, deep-space telemetry, and orbital telescope positioning.

❖ The Identified Gap: Theoretical vs. Dynamic Disparity

A critical review of existing literature reveals a significant reliance on idealized mathematical models to predict maneuver performance. Most references prioritize the conservation of angular momentum equations but fail to account for the non-linearities and parasitic effects inherent in a physical mechanism. Specifically, there is a notable absence of integrated multi-body dynamics (MBD) and structural analysis to validate how mechanical imbalances, bearing friction, and structural deformations affect the actual slew maneuver accuracy.

❖ Problem Statement

There is a need to develop a comprehensive design-to-validation framework for a compact reaction wheel system that transcends purely theoretical estimates. While the basic physics of a 30°maneuver can be calculated manually, the actual system performance is constrained by:

- Structural Integrity: How the flywheel responds to high centrifugal loads (ANSYS).
- Dynamic Fidelity: How motor torque ripples and internal friction impact settling time and jitter (ADAMS).
- Geometric Optimization: How to achieve maximum inertia within the strict mass and volume envelopes of a nano-satellite.

❖ Technical Objectives

The objective of this project is to synthesize a robust design and analysis pipeline to bridge the gap between theory and hardware:

- Synthesis: Perform high-fidelity theoretical calculations to size the flywheel (Radius, Mass, and Thickness) based on target slew rates and motor RPM limits.
- Virtual Prototyping: Construct a precise CAD model in Creo Parametric to define the system's mass properties and center of gravity.
- Structural Validation: Conduct Finite Element Analysis (FEA) in ANSYS to ensure the mechanism can withstand rotational stresses without deformation.

4. DATA COLLECTION

- Overall System Architecture

1. Outer Body Structure (Satellite Frame)

The outer structure of the system is designed as a rigid cubic frame representing the satellite body. It is fabricated using six square plates of mild steel (MS), each of thickness 3 mm.

Key Features:

- Material: Mild Steel (MS)
- Thickness: 3 mm
- Configuration: Closed cube (6 faces)
- Function:
    - Provides structural rigidity
    - Houses all internal components (reaction wheels, electronics, motor mounts)
    - Acts as the main reference body for attitude motion

Reason for Selection:

- Mild steel is chosen due to its:
    - Good strength and stiffness
    - Ease of machining and welding
    - Cost-effectiveness for prototyping

2. Reaction Wheel Configuration

The system uses three reaction wheels, each aligned along mutually perpendicular axes (X, Y, Z directions) to achieve full 3-axis attitude control.

Wheel Specifications:

- Material: Mild Steel
- Thickness: 2 mm
- Diameter: 52 mm
- Number of Wheels: 3
- Orientation: Orthogonal (X, Y, Z axes)

Function:

- Each wheel controls rotation about one axis
- By accelerating/decelerating the wheel, torque is generated on the satellite body

3. Reaction Wheel Assembly

Each reaction wheel is mounted on a BLDC motor shaft and forms a compact actuator unit. The assembly is designed for proper alignment, balance, and secure mounting.

Components of Assembly:

1. Reaction Wheel (Flywheel)
    a. Mounted directly on motor shaft
    b. Provides required inertia for torque generation
2. Threaded Shaft
    a. Transfers torque from motor to wheel
    b. Ensures rigid connection
3. Collar
    a. Maintains axial positioning of wheel
    b. Prevents unwanted axial movement
4. Nut (Top Locking Element)
    a. Secures the wheel firmly to shaft
    b. Prevents loosening during high-speed rotation
5. BLDC Motor
    a. Drives the reaction wheel
    b. Provides controlled angular velocity
6. Mount Plate
    a. Fixes motor to satellite structure
    b. Ensures alignment and stability

4. BLDC Motor Specifications

The system uses an A2212 1000KV BLDC motor for each reaction wheel.

Electrical Specifications:

- KV Rating: 1000 RPM/V
- Operating Voltage: 7.4V – 11.1V (2S–3S LiPo)
- Max Current: ~12 A
- Power Output: ~150 W

Mechanical Specifications:

- Weight: ~55–60 g
- Shaft Diameter: 3.17 mm
- Motor Diameter: ~27–28 mm
- Motor Height: ~30 mm

Reason for Selection:

- High speed capability
- Lightweight and compact
- Easily available and cost-effective
- Suitable for rapid prototyping

5. Electronics and Control Components

The system includes several electronic components for sensing, control, and actuation.

List of Components and Their Functions:

1. Microcontroller (e.g., Arduino / STM32)

- Acts as the main controller
- Executes control algorithms (PID)
- Processes sensor data and generates control signals

2. Motor Driver (ESC – Electronic Speed Controller)

- Controls speed of BLDC motor
- Converts PWM signal into motor rotation
- Enables precise speed control

3. IMU Sensor (Gyroscope + Accelerometer)

- Measures angular velocity and orientation
- Provides real-time feedback for control system

4. Power Supply System

- Typically Li-Po battery
- Supplies power to:
    - Motors
    - Controller
    - Sensors

5. Voltage Regulator

- Steps down battery voltage
- Provides stable voltage to electronics

6. Connecting Wires and PCB

- Ensures electrical connections
- Organized layout of circuit

7. Mounting Hardware

- Screws, bolts, spacers
- Used for fixing components inside structure

6. Working Integration of Components

- The IMU sensor detects current orientation

- Data is sent to the microcontroller
- Controller computes required correction torque
- Signal is sent to ESC
- ESC drives the BLDC motor
- Motor rotates the reaction wheel
- Due to conservation of angular momentum, the satellite body rotates in the opposite direction.

5. DESIGN AND ANALYSIS

- Reaction Wheel Design

These are the steps used in the design of reaction wheel parameters:

The complete vector equation of rotational motion for the satellite system which is basically angular momentum conservation equation expanded.

$$T_{ext} = I_S \times \dot{\omega}_S + \omega_S \times I_S \times \omega_S + I_{RW} \times \dot{\omega}_{RW} + \omega_S \times I_{RW} \times \omega_{RW}$$

To make analysis simpler, it is decided that only one wheel will rotate at a time, thus making the gyroscopic terms zero.

$$T_{ext} = I_S \times \dot{\omega}_S + I_{RW} \times \dot{\omega}_{RW}$$

In absence of external torque,

$$I_S \times \omega_S = I_{RW} \times \omega_{RW}$$

Moment of inertia of satellite is taken from CAD using iterative methods. I_S is taken to be the maximum of (Ixx, Iyy, Izz), which is Iyy = 0.1478619 kgm^2.

Slew rate or cruising speed (ω_S) can be chosen 2°/sec to 5°/sec. Let us take ω_S = 5°/sec.

The operational speed of the motor is between 8000-12000 rpm. Taking $\omega_{RW} = \omega_{max}$ at 9000 rpm.

$$I_S \times \omega_S = I_{RW} \times \omega_{RW}$$

$$0.1478619 \times (5 \times \pi/180) = I_{RW} \times (2\pi \times 9000/60)$$

$$I_{RW} = 1.36909 \times 10^{-5} \text{ kgm}^2$$

Fixing the thickness of reaction wheel to be 2mm and density of Alloy Steel to be 7850 kg/m^3 to find the required radius of reaction wheel.

$$I_{RW} = ½ \times M \times R^2$$

$$I_{RW} = ½ \times (\rho \times \pi R^2 \times t) \times R^2$$

$$1.36909\times10^{-5} = \tfrac{1}{2} \times 7850 \times \pi \times R^4 \times 0.002$$

$$R = 0.027296 \text{ m} = 27.296 \text{ mm}$$

$$M = 0.036750 \text{ kg} = 36.750 \text{ gm}$$

The calculations done in MS Excel for varies iterations in parameters

I_s (kgm^2)	ω_S (°/s)	ω_S (rad/s)	RPM	ω_{RW} (rad/s)	I_{RW} (kgm^2)	t_{RW} (mm)	R_{RW} (mm)	M_{RW} (gm)
0.14786	2	0.03490	9000	942.480	5.476×10^{-6}	2	21.708	23.243
0.14786	2	0.03490	5000	523.600	9.857×10^{-6}	2	25.144	31.183
0.14786	2	0.03490	9000	942.480	5.476×10^{-6}	2	21.708	23.243
0.14786	5	0.08726	9000	942.480	1.369×10^{-5}	2	27.296	36.750
0.14786	5	0.08726	5000	523.600	2.464×10^{-5}	2	31.617	49.305
0.14786	5	0.08726	9000	942.480	1.369×10^{-5}	2	27.296	36.750

Out of these calculations, the one illustrated above is the 4th row of the table.

❖ CAD Model

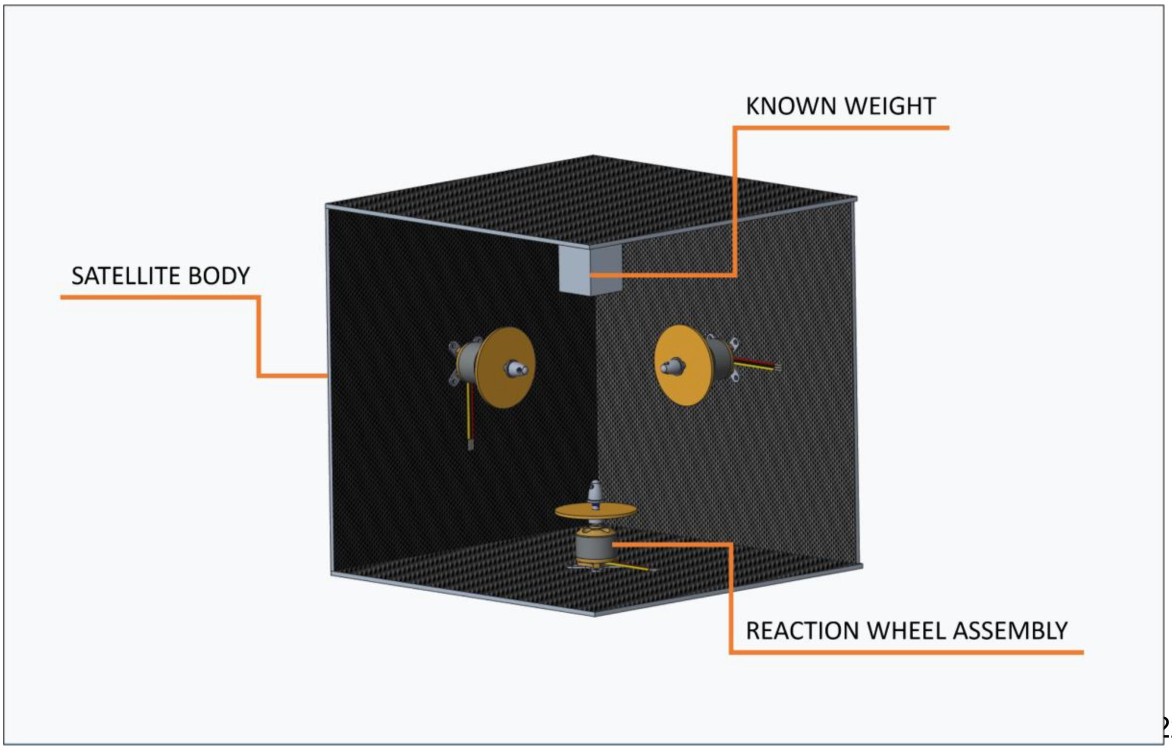

Figure 1CAD MODEL 1.1

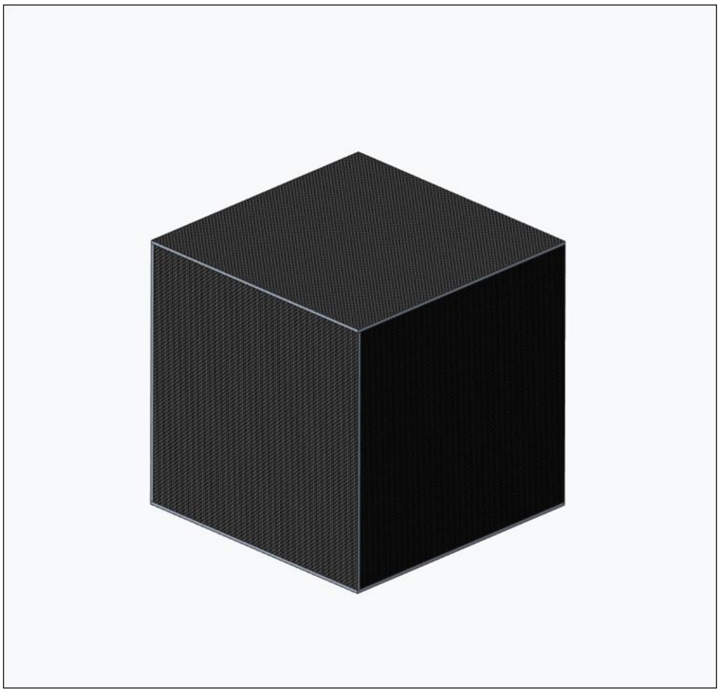

Figure 2 CAD MODEL 1.2

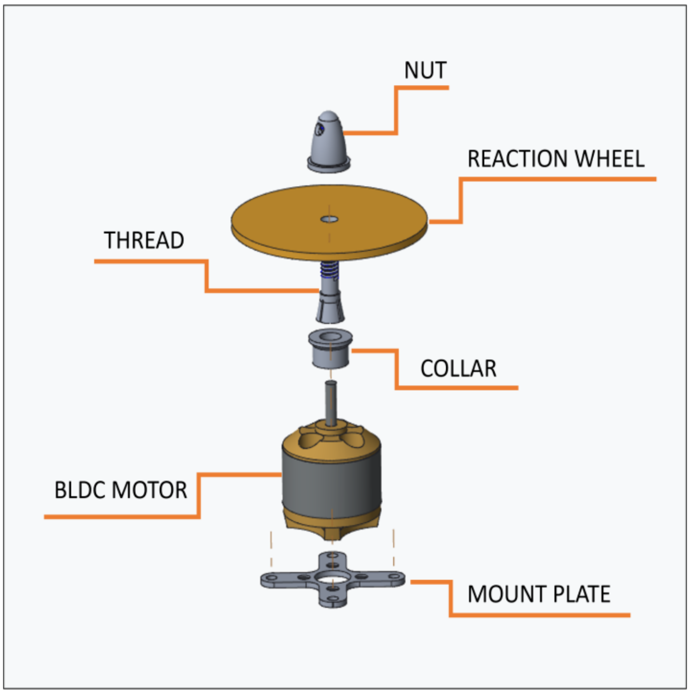

Figure 3 CAD MODEL 1.3

❖ MBD ANALYSIS

KINEMATIC ANALYSIS OF REACTION WHEEL ATTITUDE CONTROL SYSTEM (RWACS) IN MSC ADAMS

1. Introduction

The objective of this study is to perform a kinematic analysis of a reaction wheel system using MSC ADAMS, focusing on the relationship between motor input and satellite angular response. The simulation helps in understanding system dynamics such as angular displacement, velocity, acceleration, torque generation, and electrical characteristics like current consumption.

2. Geometry and Model Description

The model consists of the following main components:

- Satellite Body (Main Frame):

    - Represented as a rigid body with defined mass and moment of inertia.
    - Free to rotate about a specified axis.
- Reaction Wheel (Flywheel):
    - Mounted internally on the satellite body.
    - Designed as a rotating disc with known mass and radius.
    - High moment of inertia to maximize angular momentum exchange.
- Motor Shaft Connection:
    - The reaction wheel is connected to the satellite via a revolute joint.
    - This joint allows rotation about a single axis (typically Z-axis).
- Reference Frame:
    - A global reference frame is defined.
    - Satellite motion is measured relative to this frame.

Assumptions:

- System is considered rigid (no deformation).
- External disturbances (gravity gradient, drag, etc.) are neglected.
- Only internal torque exchange is considered.

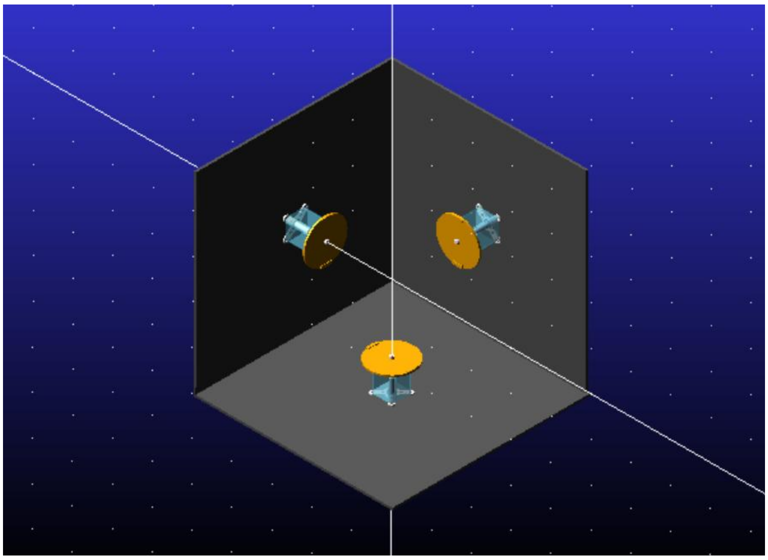

Figure 4 ADAMS MODEL

3. Boundary Conditions and Loading

Boundary Conditions:

- The satellite body is either:
    - Free in space (floating condition)
- The reaction wheel is connected via a revolute joint, allowing:
    - Rotation about one axis only.

Initial Conditions:

- Initial angular velocity of both satellite and wheel = 0 rad/s
- Initial angular displacement = 0°

Loading Conditions:

- No external forces or torques are applied.
- Only internal torque generated by motor drives the system.

4. Motor Creation and Settings in MSC ADAMS

The motor is the key driver of the system. It is applied at the revolute joint connecting the reaction wheel and satellite.

Steps to Create Motor:

1. Input Function:

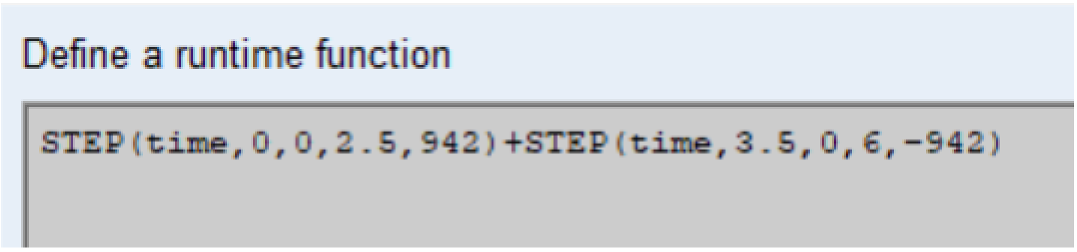

Figure 5 ADAMS INPUT FUNCTION

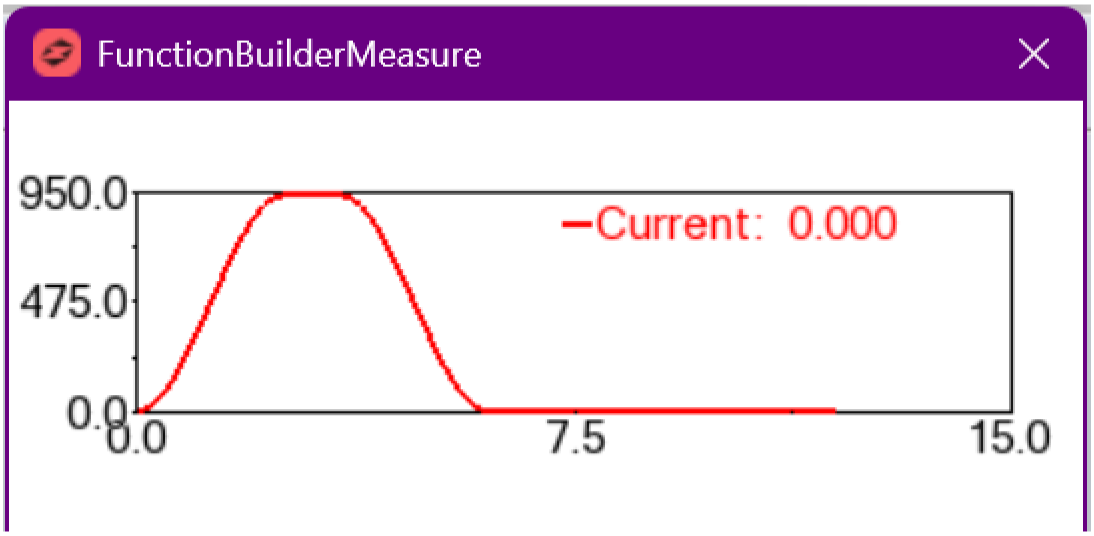

Figure 6 ADAMS INPUT FUNCTION GRAPH

2. Motor Profile:

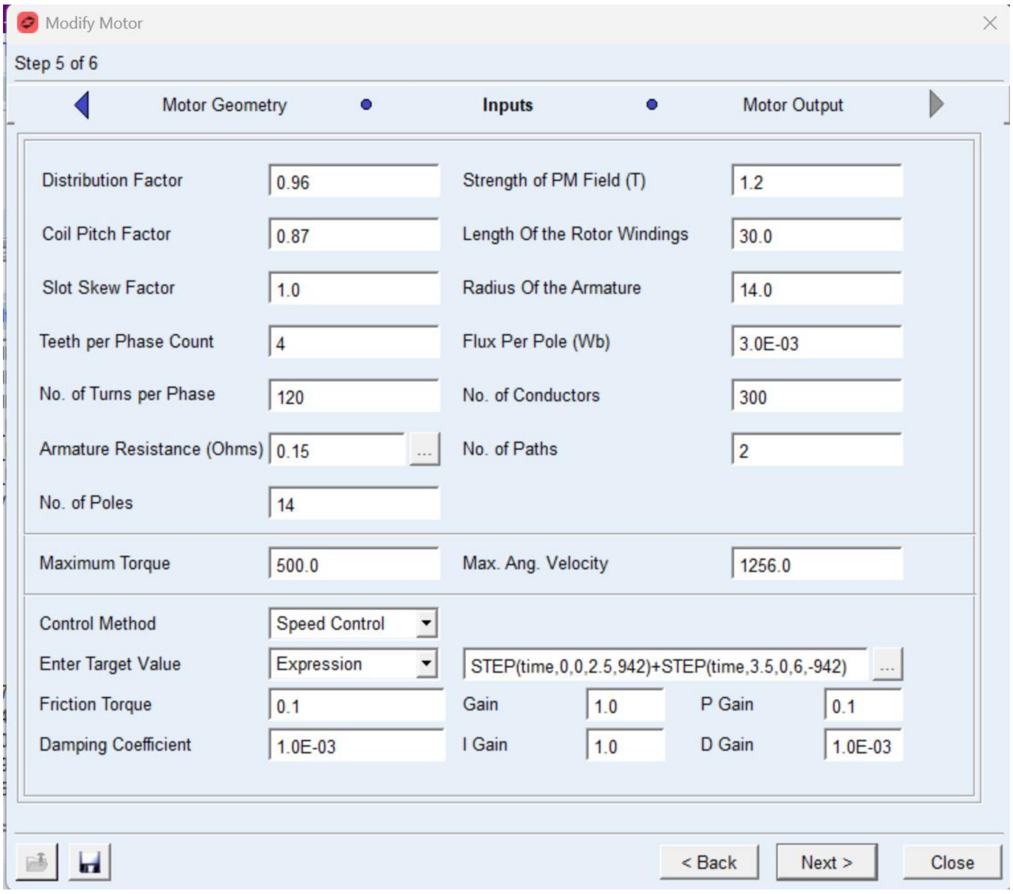

Figure 7 ADAMS MOTOR SPECS

3. Units:
    - Ensure consistency (rad/s preferred for analysis).

Motor Parameters Considered:

- Maximum angular velocity (e.g., up to 9000–12000 RPM)
- Torque limits based on motor capability
- Time required to reach desired speed

5. Simulation Outputs and Results

The following parameters were plotted and analyzed:

1. Motor RPM vs Time

- Shows how quickly the reaction wheel accelerates.
- Helps verify motor capability vs required speed.

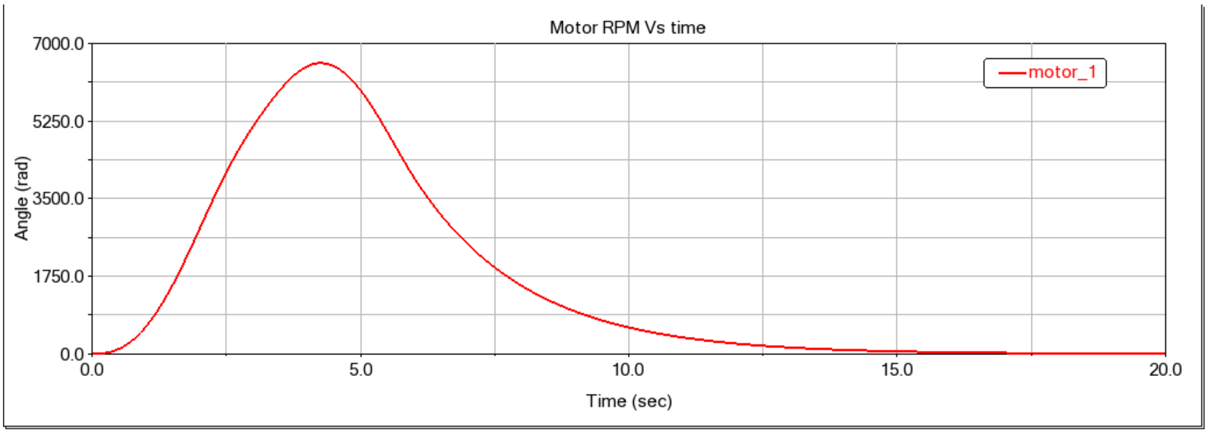

Figure 8 MOTOR RPM vs TIME

2. Angular Velocity (Wheel & Satellite)

- Reaction wheel velocity increases.
- Satellite rotates in opposite direction due to conservation of momentum.

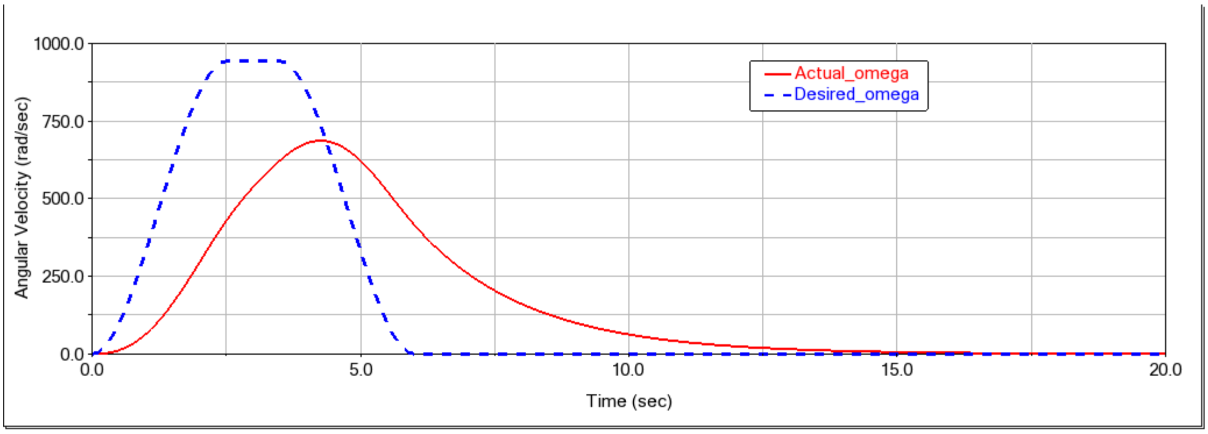

Figure 9 ANGULAR VELOCITY vs TIME

3. Angular Acceleration

- Peaks during motor startup.
- Indicates torque demand.

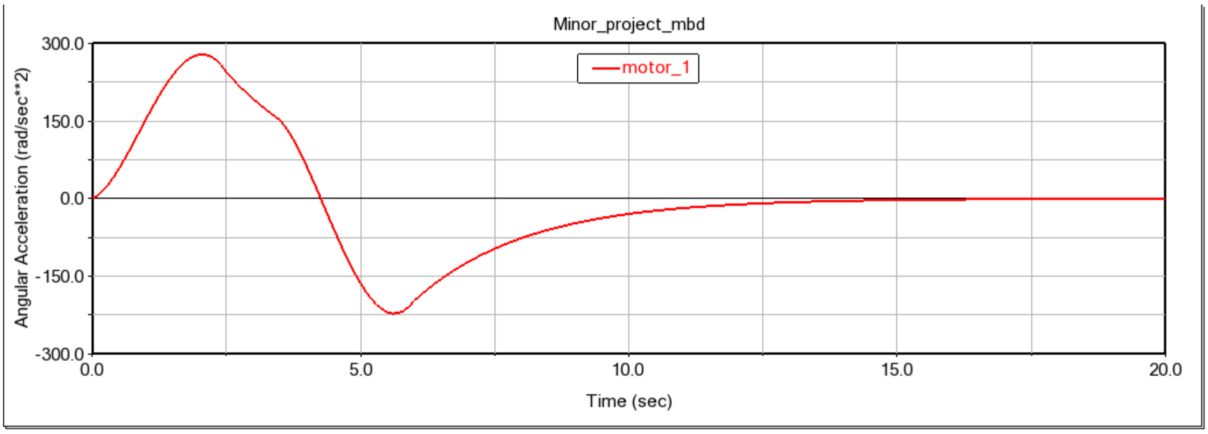

Figure 10 ANGULAR ACC. vs TIME

4. Torque vs Time

- High during acceleration phase.
- Drops during constant velocity.
- Reverses during deceleration (if applied).

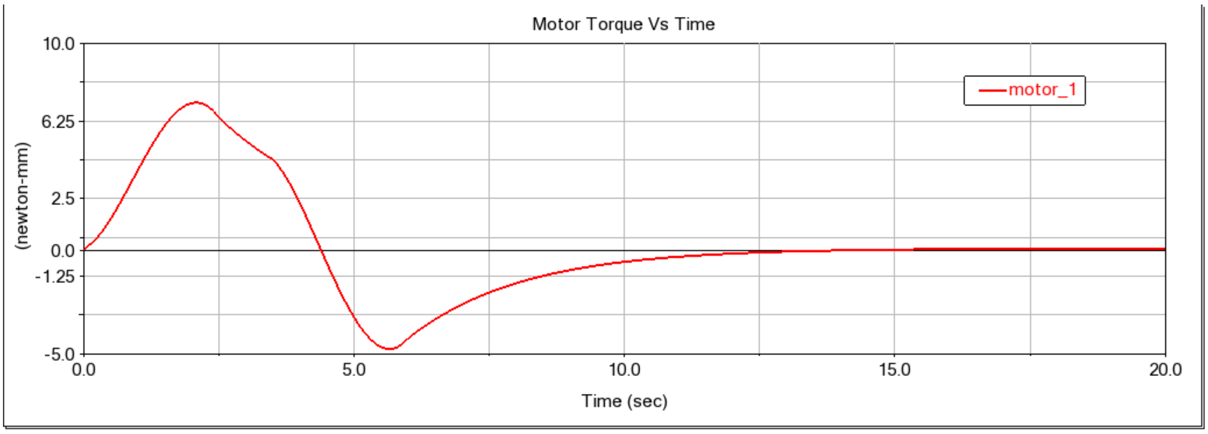

Figure 11 TORQUE vs TIME

5. Current Drawn

- Directly proportional to torque.
- Peaks during acceleration phase.
- Important for power system design.

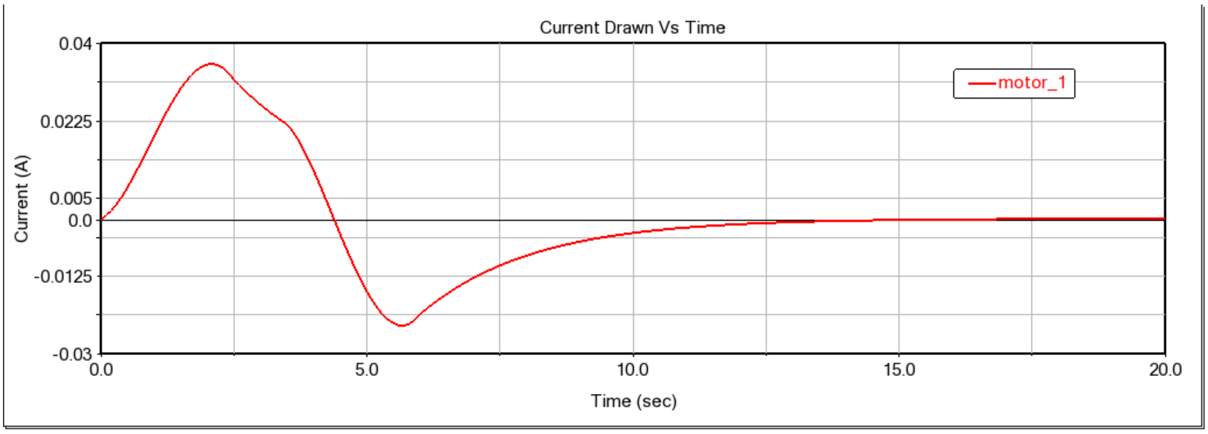

Figure 12 CURRUNT DRAWN vs TIME

6. Satellite Angle vs Time

- Key output for attitude control.
- Confirms whether desired rotation (e.g., 30° in 11 sec) is achieved.

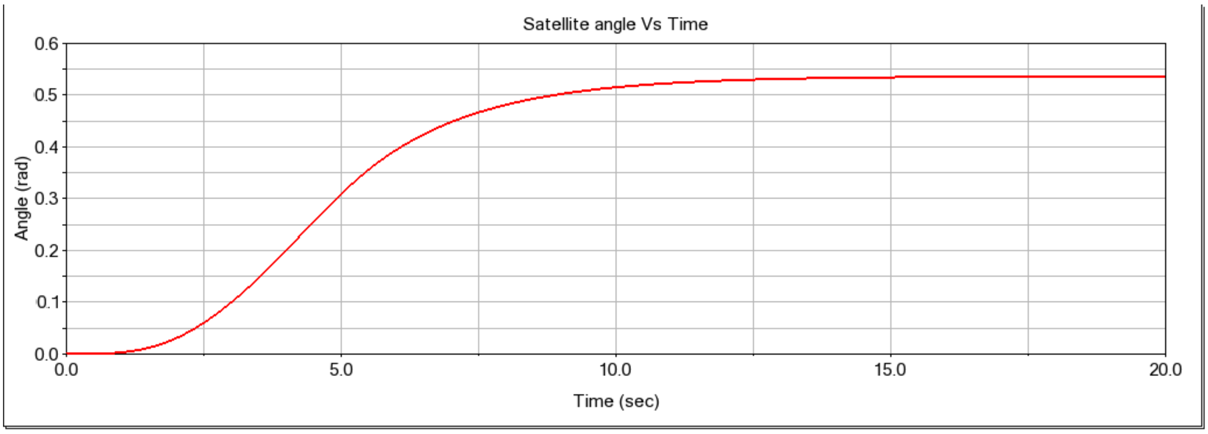

Figure 13 SATELLITE ANGLE vs TIME

❖ CAE ANALYSIS

CAE ANALYSIS OF REACTION WHEEL ATTITUDE CONTROL SYSTEM (RWACS)

1. Introduction

Computer-Aided Engineering (CAE) analysis was performed to validate the structural integrity and dynamic behavior of the Reaction Wheel Attitude Control System (RWACS). The primary objective of this analysis is to ensure that the designed components, particularly the flywheel and housing, can safely withstand operational and launch-induced loads without failure.

Special emphasis was given to:

- Avoidance of resonance with motor excitation frequency
- Structural stability under random vibration (launch conditions)
- Verification of design reliability using standardized aerospace loading conditions

2. Geometry and Model Description

The CAD model of the reaction wheel assembly was developed in Creo Parametric and imported into ANSYS Mechanical for analysis.

The CAE study includes:

- Flywheel (primary rotating component)
- RWACS housing (supporting structure)

The flywheel is designed as a thin circular disc with a central hole, based on calculated parameters:

- Radius: 27.29 mm
- Thickness: 2 mm
- Mass: 36.75 g

The RWACS housing encloses the motor and flywheel assembly and provides mounting interfaces to the satellite structure.

3. Material Properties

The material selected for the flywheel is Alloy Steel. The following properties were used in the simulation:

Property	Value
Density	7850Kg/m^3
Young's Modulas	2 X 10^5 Mpa
Poisson's Ratio	0.3

The material is assumed to be linear elastic and isotropic for all simulations.

4. Meshing Details

The geometry was discretized using tetrahedral elements in ANSYS Mechanical.

Mesh Specifications:

- Element Type: Hex & Tetrahedral
- Mesh Method: Patch conforming
- Refinement applied at:
    - Inner hole region
    - Outer circumference

Mesh Quality Criteria:

- Skewness < 0.6
- Orthogonality: 0.5 – 0.8
- Tetrahedral Collapse: 0.7 – 1

These criteria ensure accurate stress distribution and numerical stability of the solution.

A fine mesh was used in critical regions to capture stress concentration and dynamic response accurately

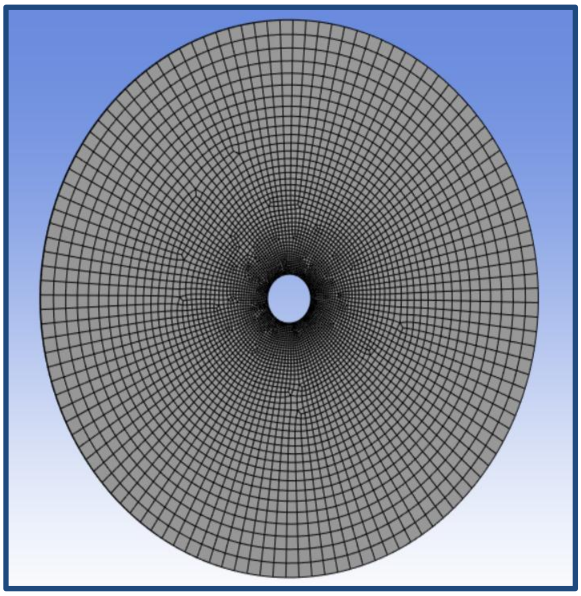

Figure 15 FLYWHEEL MESH

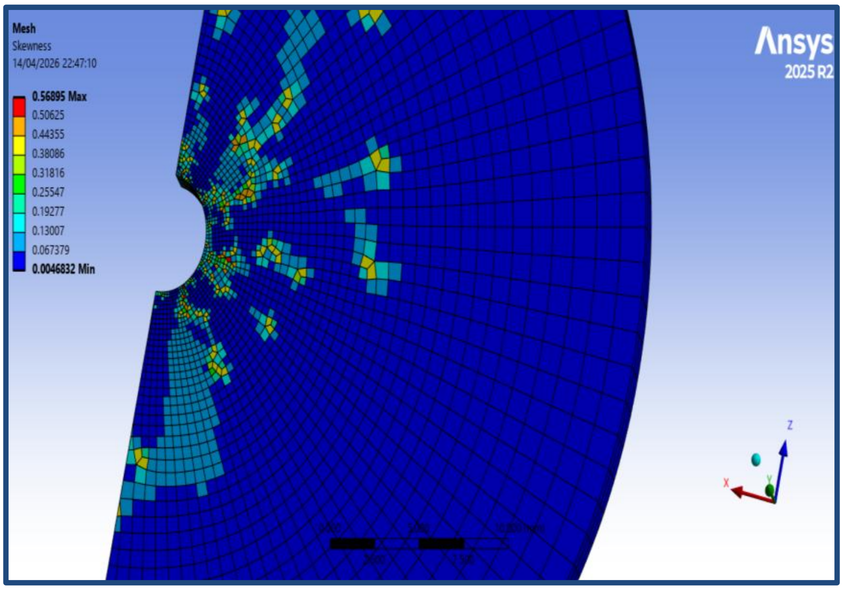

Figure 14 SKEWNESS CHECK

5. Boundary Conditions and Loading

5.1 Modal Analysis (Pre-Stressed)

To simulate realistic operating conditions, a pre-stressed modal analysis was performed.

- The mounting holes/regions were assigned fixed support, restricting all degrees of freedom

- A rotational velocity corresponding to the operating speed (9000 RPM) was applied about the axis of rotation

This approach accounts for centrifugal stiffening effects in the rotating flywheel.

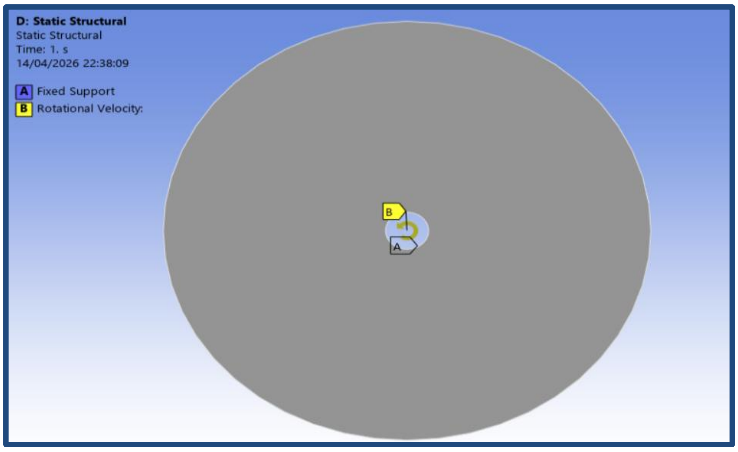

Figure 16 BOUNDARY CONDITION FOR PRE-STRESS MODAL

Mode	Frequency(Hz)
1	2074.5
2	2074.5
3	2678.7
4	3553.8
5	3553.8
6	4595.9

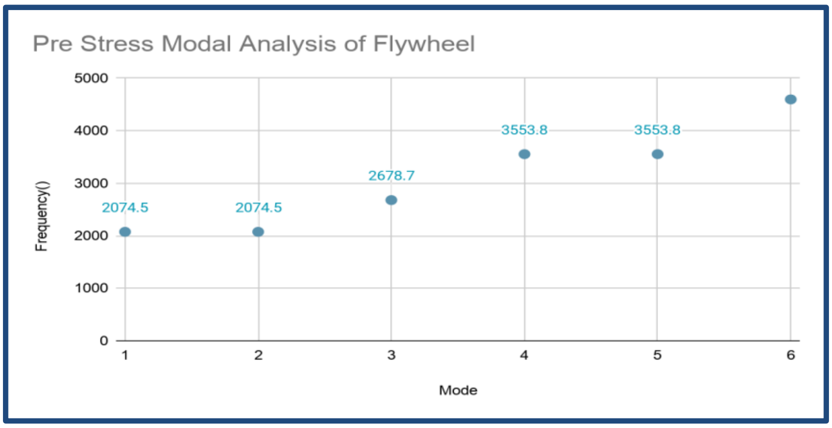

Figure 17 PRE STRESS MODEL MODAL ANALYSIS

Table- Frequency of first 6 Modes

5.2 Random Vibration Analysis

Random vibration analysis was conducted to simulate launch and environmental conditions.

- Input: Power Spectral Density (PSD) acceleration curve
- Source: General Environmental Verification Specification (GEVS)
- Frequency Range: (e.g., 20 Hz – 2000 Hz)

The PSD input represents realistic aerospace vibration conditions experienced during rocket launch.

The same PSD curve was consistently used for both:

- Flywheel

- RWACS housing

to ensure uniform excitation conditions across all components.

Figure 18 PSD G ACCELERATION (G^2/HZ) CURVE FOR INPUT

6. Solver Settings

- Software: ANSYS Mechanical
- Analysis Types:
    - Pre-stressed Modal Analysis
    - Random Vibration Analysis (PSD)

The modal results were used as input for the random vibration analysis to evaluate system response under dynamic excitation.

6. IMPLEMENTATION

1. Introduction

- The Reaction Wheel Attitude Control System (RWACS) is an effective method for controlling the orientation of nano satellites without using propellant. It operates on the principle of conservation of angular momentum, where variation in reaction wheel speed produces an equal and opposite rotation of the satellite. This enables precise and continuous attitude control required for communication alignment and payload orientation.

2. System Implementation

- The RWACS was implemented through integrated mechanical design, modeling, and control system development. The reaction wheel assembly, including flywheel, motor housing, and mounting structure, was designed using CAD software with emphasis on balance, compactness, and vibration reduction.
- System parameters such as moment of inertia, required torque, and angular velocity limits were calculated analytically. A dynamic model based on rigid body dynamics was developed to describe the interaction between the satellite and reaction wheels.
- The 3D model was imported into MSC ADAMS for dynamic simulation, where multiple iterations were performed to optimize system performance.
- Structural validation was carried out in ANSYS through modal analysis, random vibration analysis, and housing strength evaluation to ensure reliability under operational and launch conditions.

3. Experimentation

- Simulation-based experimentation was conducted to evaluate system performance under step inputs, disturbances, and multi-axis motion.The results indicate that the RWACS provides accurate, stable, and reliable attitude control suitable for nano satellite applications.

7. RESULTS AND DISCUSSION

7.1 Modal Analysis of Flywheel

The natural frequencies and corresponding mode shapes of the flywheel were extracted.

The analysis was performed as a pre-stressed modal analysis, accounting for the operational rotational effects of the flywheel. The first six natural frequencies were obtained in the range of 2000 Hz to 5000 Hz, which are significantly higher than the motor excitation frequency (200 Hz), thereby ensuring that resonance conditions are avoided during operation.

Key Observation:

- Motor operating speed = 9000 RPM
- Excitation frequency = 9000 / 60 = 200 Hz

Since none of the natural frequencies coincide with 200 Hz, the design is safe from resonance.

7.2 Random Vibration Analysis of Flywheel

The response of the flywheel under random vibration loading was evaluated using PSD analysis.

Result:

- Output: Frequency vs Displacement PSD
- Maximum displacement observed within acceptable limits
- No peak observed near excitation frequency (200 Hz)

The displacement results were evaluated at the mounting/interface locations, representing the most critical regions for structural integrity.

Conclusion:

The flywheel exhibits stable dynamic behavior under random vibration loading and satisfies safety requirements.

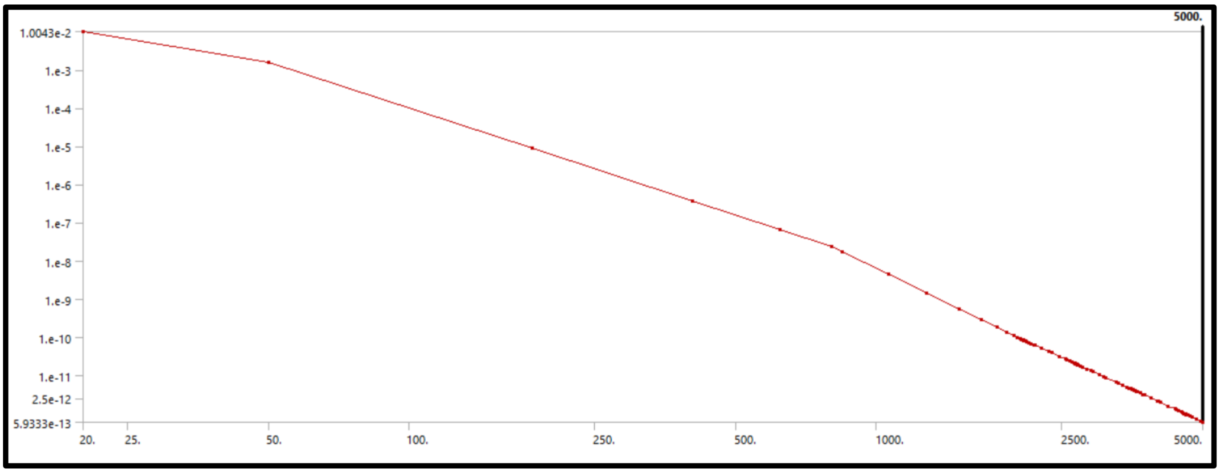

Figure 19 FREQUENCY(X-AXIS) V/S DISPLACEMENT(Y-AXIS) FOR FLYWHEEL

7.3 Random Vibration Analysis of RWACS Housing

The RWACS housing was analyzed under the same PSD loading conditions.

Results:

- Displacement PSD response remains within safe limits
- No resonance observed near excitation frequency
- Strain contours indicate uniform distribution without localized failure

The contour plots of normal elastic strain confirm that the structure can withstand the applied vibration loads without structural damage.

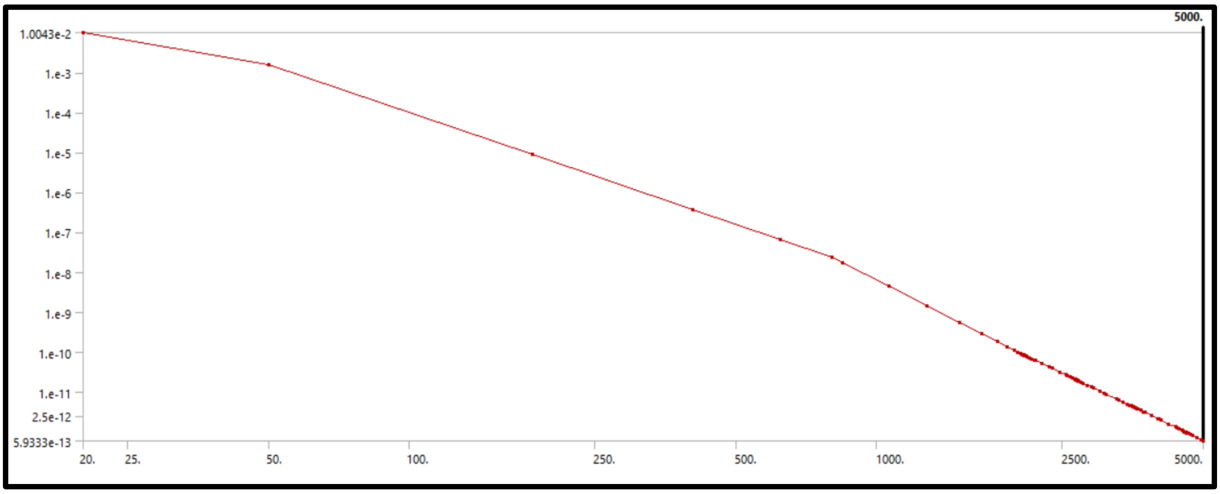

Figure 20 FREQUENCY(X-AXIS) V/S DISPLACEMENT(Y-AXIS) FOR FLYWHEEL

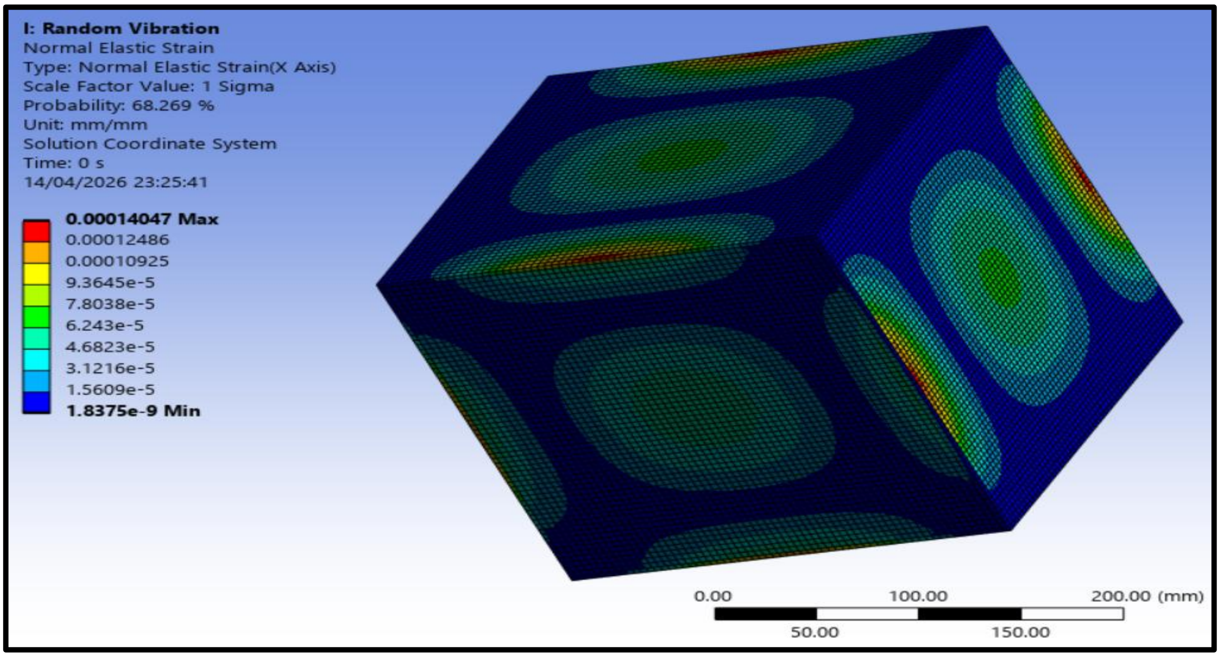

Figure 21 RANDOM VIBRATION RESPONSE SHOWING CONTOUR OF NORMAL ELASTIC STRAIN (X-DIRECTION) OF THE STRUCTURE UNDER GEVS PSD LOADING.

MBD Results -

- Max Current drawn = 0.035 Amp
- Max Angular acceleration = 278.1 rad/sec sq.
- Desired Max. Angular Velocity = 942 rad/sec
- Actual Max. Angular Velocity = 686 rad/sec
- Max RPM = 6550
- Max Torque = 7 Nmm
- The Satellite covers 0.52 rad (30°) in 11sec

8. CONCLUSIONS AND SCOPE OF FUTURE

The implementation and experimentation of the Reaction Wheel Attitude Control System (RWACS) demonstrate its effectiveness as a reliable and efficient solution for attitude control in nano satellites. By utilizing the principle of conservation of angular momentum, the system enables precise orientation control without the need for propellant, making it highly suitable for small satellite missions where mass and power constraints are critical.

The integration of mechanical design, analytical parameterization, and dynamic modeling ensured a strong theoretical foundation for system development. The use of advanced tools such as Creo Parametric, MSC ADAMS, and ANSYS enabled comprehensive design validation through multi-body dynamics simulation and structural analysis. Modal and random vibration analyses confirmed the structural integrity of the system under operational and launch conditions.

Simulation-based experimentation verified the system's ability to achieve stable and accurate attitude control under various conditions, including disturbances and multi-axis motion. Performance metrics such as settling time, overshoot, and steady-state error indicated satisfactory system response and robustness.

Overall, the study confirms that RWACS is a viable and scalable solution for nano satellite applications. The methodology adopted in this work provides a systematic approach for the design, analysis, and validation of attitude control systems, and can be extended to more advanced control strategies and real-time hardware implementation in future developments.

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